

Document number:	P2697R1
Date:	2023-06-15
Project:	Programming Language C++
Audience:	LWG
Reply-to:	Michael Florian Hava ¹ < mfh.cpp@gmail.com >

Interfacing bitset with string_view

Abstract

This paper proposes amending the interface of `bitset` to support construction from `basic_string_view`.

Tony Table

Before		Proposed	
<code>bitset b0{""};</code>	✓	<code>bitset b0{""};</code>	✓
<code>bitset b1{""sv};</code>	✗	<code>bitset b1{""sv};</code>	✓
<code>bitset b2{""s};</code>	✓	<code>bitset b2{""s};</code>	✓
<code>//concerning LWG2946</code>		<code>//concerning LWG2946</code>	
<code>bitset b3{("", 1)};</code>	✗	<code>bitset b3{("", 1)};</code>	✗

Revisions

R0: Initial version

R1: Updates after LWG Review on 2023-06-14:

- Modified wording according to LWG guidance.

Motivation

[\[string.view\]](#) specifies `basic_string_view`, a vocabulary type template that represents an immutable reference to some string-like object. Unless a string can be moved from source to target, it is generally advisable to pass "immutable stringy inputs" by `basic_string_view`. Doing so obviates the need for multiple overloads and enables support for user-defined types.

[\[template.bitset\]](#) specifies the class templates `bitset` to represent a fixed size sequence of bits. It can be initialized from the biggest fundamental unsigned type (`unsigned long long int`) and a string. As `bitset` predates the introduction of `basic_string_view`, it only supports construction from strings of two forms: `const CharT *` and `basic_string<CharT, Traits, Allocator>`, with `CharT`, `Traits` and `Allocator` being deduced in the respective constructor and then promptly discarded, as `bitset` is independent of these types.

This leads to an embarrassing problem when following the aforementioned recommendation: Every `basic_string_view` must either be:

- converted to a temporary `basic_string`, introducing an unnecessary(!) copy as `bitset` only reads from the string for initialization, or

¹ RISC Software GmbH, Softwarepark 32a, 4232 Hagenberg, Austria, michael.hava@risc-software.at

- extracted (via `.data()`); This approach places additional burden on the user as the respective `const CharT *` may not be `\0`-terminated and additional constructor parameters need to be provided to prevent an out of bounds access.

This paper aims to solve these issues by introducing direct support for `basic_string_view`.

Design space

This paper proposes to add a new constructor taking a `basic_string_view` to `bitset`. Contrary to other extensions to similar overload sets (e.g. [P2495](#)), [LWG2946](#) does not apply here as all existing constructors of `bitset` are explicit.

Impact on the Standard

This proposal is a pure library addition. One existing standard library class is modified in a non-ABI-breaking way. Overload resolution for existing code is not affected by the introduced overload.

Implementation Experience

The proposed overload set has been implemented on [\[https://godbolt.org/z/56aaE3qP7\]](https://godbolt.org/z/56aaE3qP7) for evaluation.

Proposed Wording

Wording is relative to [\[N4917\]](#). Additions are presented like **this**, removals like **this**.

[\[version.syn\]](#)

In [\[version.syn\]](#), add:

```
#define cpp_lib_bitset_YYYYMMML //also in <bitset>
```

Adjust the placeholder value as needed to denote this proposal's date of adoption.

[\[template.bitset.general\]](#)

In [\[template.bitset.general\]](#), in the synopsis, add the proposed overload:

```
...
// 22.9.2.2, constructors
constexpr bitset() noexcept;
constexpr bitset(unsigned long long val) noexcept;
template<class charT, class traits, class Allocator>
constexpr explicit bitset(
    const basic_string<charT, traits, Allocator>& str,
    typename basic_string<charT, traits, Allocator>::size_type pos = 0,
    typename basic_string<charT, traits, Allocator>::size_type n = basic_string<charT, traits, Allocator>::npos,
    charT zero = charT{'0'},
    charT one = charT{'1'});
template<class charT, class traits>
constexpr explicit bitset(
    basic_string_view<charT, traits> str,
    typename basic_string_view<charT, traits>::size_type pos = 0,
    typename basic_string_view<charT, traits>::size_type n = basic_string_view<charT, traits>::npos,
    charT zero = charT{'0'},
    charT one = charT{'1'});
template<class charT>
constexpr explicit bitset(
    const charT* str,
    typename basic_string_view<charT>::size_type n = basic_string_view<charT>::npos,
    charT zero = charT{'0'},
    charT one = charT{'1'});
// 22.9.2.3, bitset operations
...
```

[bitset.cons]

In [bitset.cons]:

```
template<class charT, class traits, class Allocator>
constexpr explicit bitset(
    const basic_string<charT, traits, Allocator>& str,
    typename basic_string<charT, traits, Allocator>::size_type pos = 0,
    typename basic_string<charT, traits, Allocator>::size_type
        n = basic_string<charT, traits, Allocator>::npos,
    charT zero = charT('0'),
    charT one = charT('1'));
template<class charT, class traits>
constexpr explicit bitset(
    basic_string_view<charT, traits> str,
    typename basic_string_view<charT, traits>::size_type pos = 0,
    typename basic_string_view<charT, traits>::size_type n = basic_string_view<charT, traits>::npos,
    charT zero = charT('0'),
    charT one = charT('1'));
3     Effects: Determines the effective length rlen of the initializing string as the smaller of n and str.size() - pos. Initializes the first M bit
    positions to values determined from the corresponding characters in the string str. M is the smaller of N and rlen.
...
template<class charT>
constexpr explicit bitset(
    const charT* str,
    typename basic_string<view<charT>::size_type n = basic_string<view<charT>::npos,
    charT zero = charT('0'),
    charT one = charT('1'));
8     Effects: As if by:
        bitset(n == basic_string<view<charT>::npos
            ? basic_string<view<charT>(str)
            : basic_string<view<charT>(str, n),
            0, n, zero, one)
```

Acknowledgements

Thanks to [RISC Software GmbH](#) for supporting this work. Thanks to Peter Kulczycki for proof reading.